1. (a) How do standards support interoperability and consistency in software projects? (CO3) [3]

(b) Explain in detail the unified process model. Compare it with iterative enhancement model. (CO1) [4]

(c) What do you understand by total quality management (TQM)? What are the advantages of TQM? Does ISO 9000 standard aim for TQM? (CO4) [3]

2. (a) Discuss the key quality attributes of software. How do they affect the overall software engineering process? (CO4) [2]

(b) Distinguish between software verification and software validation. Can one be used in place of the other? Justify your answer? (CO4) [3]

(c) Draw control flow graph and calculate cyclomatic complexity for the given code.

```
{ int i, j, k;
  for (i=0 ; i<=N ; i++)
     p[i] = 1;
  for (i=2 ; i<=N ; i++)
  {
   k = p[i]; j=1;
   while (a[p[j-1]] > a[k] {
     p[j] = p[j-1];
     j--; }
   }
   p[j]=k;
}
```

(CO4) [5]

3. (a) How do black-box and white-box testing techniques differ in terms of their approach and application? (CO4) [3]

(b) What are the basic principles of programming during the coding phase? How does structured programming help improve code readability (CO4) [4]

(c) Compare and contrast function-oriented design and object-oriented design. Discuss the advantages and limitations of both approaches. (CO3) [3]

4. (a) Define the following:
i) Actors ii) Scenario iii) Use cases iv) Class Diagram
v) Sequence Diagram (CO3) [5]

(b) What are the key architecture styles for component and connector views? Discuss their significance in system design? (CO3) [3]

(c) Is it true that whenever you increase the cohesion of your design, coupling in the design would automatically decrease? Justify your answer by using suitable examples. (CO3) [2]

5. (a) A project size of 200 KLOC is to be developed. Software development team has average experience on similar type of projects. The project schedule is not very tight. Calculate the effort, development time, average staff size and productivity of the project. (CO3) [4]

Software Project	a_b	b_b	c_b	d_b
Organic	2.4	1.05	2.5	0.38
Semidetached	3.0	1.12	2.5	0.35
Embedded	3.6	1.20	2.5	0.32

(b) Compare the advantages and disadvantages of the following two project size estimation techniques—expert judgement and Delphi technique? (CO2) [3]

(c) Define the software development process. What are the key characteristics of a good software process? (CO1) [3]